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Course Code: CSA6502

Slot: A

Course Name: Generative Ai and Large Scale Models

Course Faculty: Dr. Nithisha J

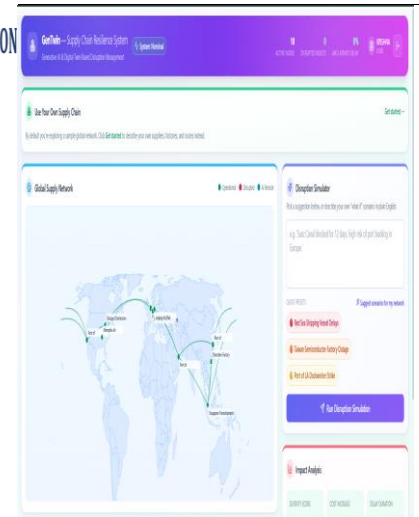
Project Title: Project GenTwin: An Generative Digital Twin for Supply Chain Disruption Simulation and Resilience Planning

Module Photographs:



MODULE-2: GENERATIVE AI - BASED DISRUPTION SIMULATION

- Generate realistic supply chain disruption scenarios using Generative AI.
- Simulate events such as supplier failures, transportation delays, and demand fluctuations.
- Analyze how each disruption affects different parts of the supply chain.
- Estimate the impact on inventory, delivery time, and operational cost.
- Identify critical risks and assign a risk score to each disruption.
- Pass the simulation results to the resilience planning module.



Project Description:

GenTwin is a Generative AI-based Digital Twin system developed to simulate supply chain disruptions and support resilience planning. It creates a virtual representation of suppliers, warehouses, inventory, transportation routes, and customer demand to model real-world supply chain operations. The system uses Generative AI to generate realistic disruption scenarios such as supplier failures, transportation delays, and demand fluctuations. It analyzes the resulting impact and provides suitable recovery strategies to improve supply chain resilience, reduce operational risks, and support better decision-making.

This module uses Generative AI to generate realistic supply chain disruption scenarios based on different operational conditions. It simulates events such as supplier failures, transportation delays, and sudden changes in customer demand to understand their effects on the supply chain. The system evaluates the impact of these disruptions on inventory levels, delivery time, operational costs, and overall performance. It also identifies critical risks, assigns risk scores, and passes the simulation results to the resilience planning module.

Student Signature

Guide Signature